MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

A
Presentation
on

"TrustChain-AI: A Decentralized Trust Intelligence Framework for Secure Autonomous AI Agent Collaboration"

By

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Outline

- Objective of Project & Literature Survey
- Problem Statement & Proposed Solution
- System Architecture & Core Working Flow
- Decentralized Identity & Trust Intelligence Lifecycle
- Adaptive Policy Enforcement & Agent Matchmaking
- Multi-Agent Collaboration & Security Operations (SOC)
- Actual Project Screenshots: Security & Swarm Operations
- Implementation Status, Key Features & Requirements
- Project Development Timeline (Gantt Chart)
- Implementation & Technical Validation
- Results Analysis & Evaluation
- Project Outcomes & Conclusion
- References

Progress Since Phase II

Area	At Phase II	At Phase III
Identity	did:key + Ed25519 demonstrated	Full did:key + W3C VC lifecycle with URDNA2015 canonicalization and Ed25519Signature2020 proofs
Trust engine	EMA scoring, 3 states + revoke	Same engine, validated with a 10-round scenario and a live /health probe
Authorization	SUSPICIOUS allowed (logged)	SUSPICIOUS blocks privileged actions (HTTP 403); human RBAC added (ADMIN, OPERATOR, VIEWER)
Discovery and matching	Not present	Explainable discovery and multi-factor MatchScore ranking
Collaboration	Not present	Bilateral sessions, sequential pipelines, mid-session abort
Storage	SQLite VDR	13-table SQLite WAL VDR with four audit streams
Frontend	Dashboard	Five-module SOC portal with Agent Investigation Drawer
Testing	40 tests	80 tests passing in 15.04 s; Vite build with 0 errors

Objective of Project

The primary objective of TrustChain-AI is to develop a decentralized trust intelligence framework that enables secure and autonomous collaboration among AI agents across multiple domains. The framework utilizes W3C Decentralized Identifiers (DIDs), Verifiable Credentials (VCs), and blockchain to establish trusted digital identities and secure cross-domain interactions.

It further introduces a Trust Intelligence Engine for continuous trust evaluation based on credential verification, reputation, and agent behaviour, while supporting privacy-preserving credential sharing, adaptive trust-based authorization. The goal is to create a scalable, interoperable, and decentralized ecosystem for reliable AI agent collaboration without relying on centralized trust authorities.

Literature Survey

Reference Paper	Research Focus	Contribution	Research Gap	TrustChain-AI Solution
AI Agents with DIDs & VCs (Base Paper)	AI Agent Identity, DID, VC	Secure decentralized AI identity & authentication	No trust scoring or reputation management	Adds Trust Intelligence Engine, dynamic trust scoring & malicious agent detection
LOKA Protocol: A Decentralized Framework for Trustworthy and Ethical AI Agent Ecosystems	AI Identity, Ethical AI	Universal AI Agent Identity Layer	No real-time trust evaluation	Continuous trust monitoring & decentralized reputation
Selective Disclosure in SSI	Privacy, Credential Sharing	Privacy-preserving credential exchange	No AI collaboration or trust analysis	Secure selective disclosure with trust evaluation
SoK: Trusting Self-Sovereign Identity	Trust Models, Security	Trust models & threat analysis	Static trust for human identities	Dynamic trust using behavior, reputation & history
Self-Sovereign Digital Identity	Decentralized Identity	Introduced Self-Sovereign Identity (SSI)	Limited to human identities	Extends SSI to autonomous AI agents using DIDs & VCs

Problem Statement

Autonomous AI agents increasingly collaborate across multiple domains, but existing systems primarily focus on identity authentication rather than continuous trust evaluation. Current approaches lack dynamic trust scoring, reputation management, malicious agent detection, and adaptive trust mechanisms, making secure and reliable agent collaboration challenging. There is a need for a decentralized trust framework that combines W3C Decentralized Identifiers (DIDs), Verifiable Credentials (VCs), blockchain, and intelligent trust evaluation to enable secure, scalable, and trustworthy collaboration among autonomous AI agents without relying on centralized authorities.

Proposed Solution

TrustChain-AI proposes a Decentralized Trust Intelligence Framework that enables secure, trustworthy, and autonomous collaboration among AI agents across multiple organizational domains. The framework integrates W3C Decentralized Identifiers (DIDs), Verifiable Credentials (VCs), and blockchain technology to provide decentralized digital identities, tamper-proof credential verification, and secure cross-domain authentication.

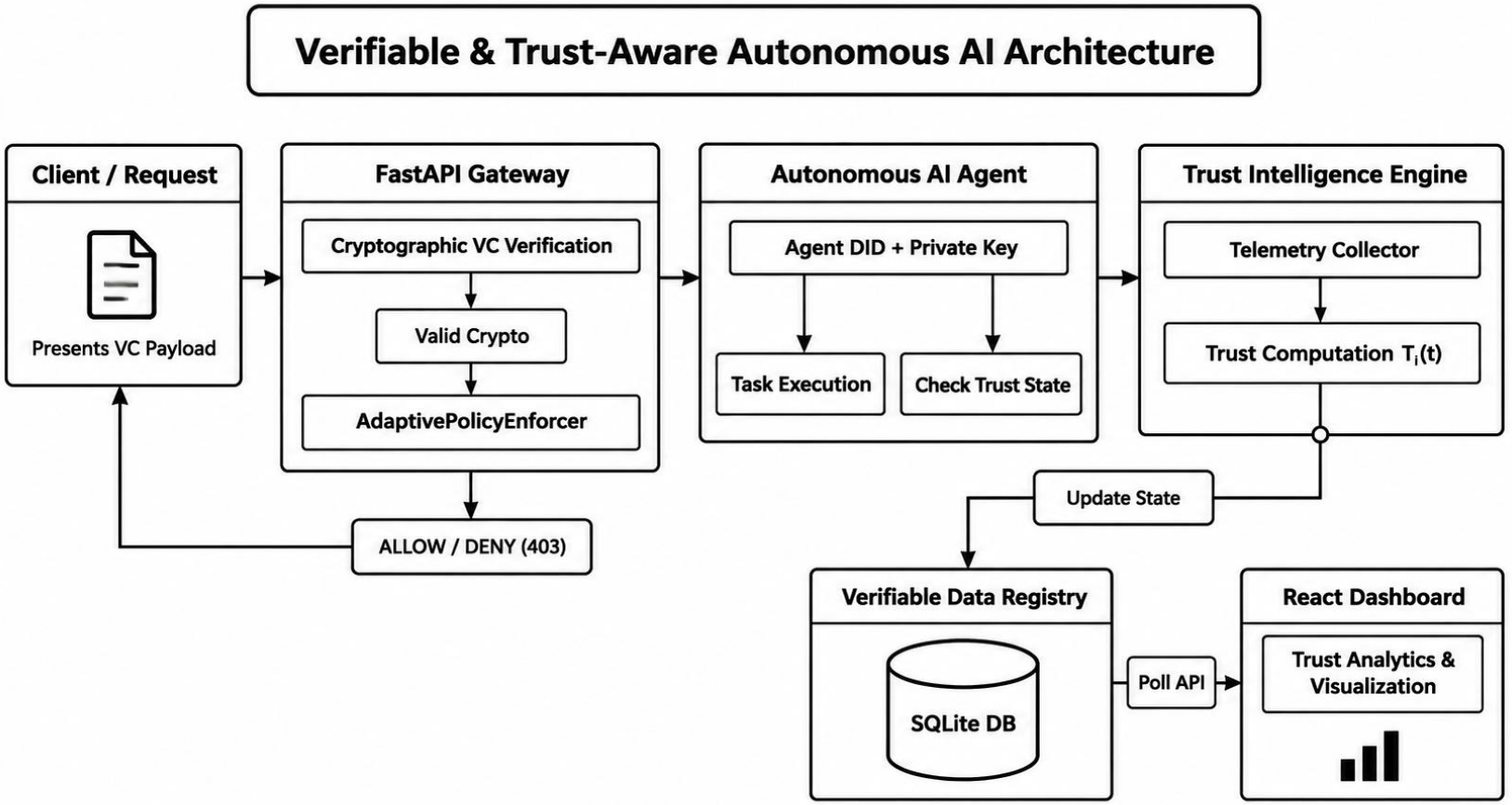
To address the limitations of existing identity-based approaches, TrustChain-AI introduces a Trust Intelligence Engine that continuously evaluates AI agents using credential verification, reputation scores, behaviour analysis, interaction history, and adaptive trust policies. The framework also supports privacy-preserving selective disclosure, malicious agent detection. ensuring secure, scalable, and future-ready collaboration without relying on centralized trust authorities.

Methodology

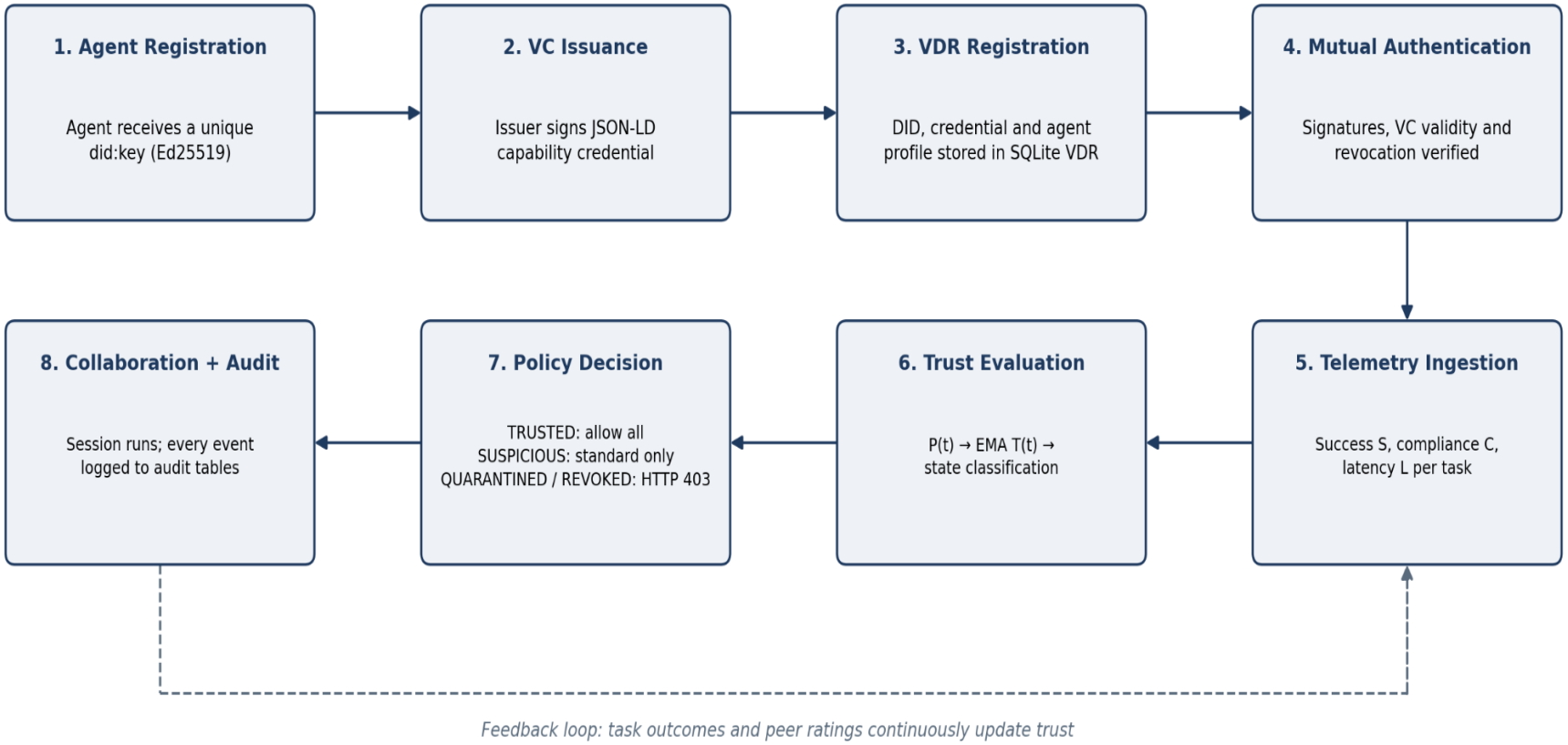
The proposed TrustChain-AI framework extends the decentralized identity model of the base research paper by integrating a Dynamic Trust Intelligence Engine for secure autonomous AI agent collaboration. Initially, each AI agent is assigned a W3C Decentralized Identifier (DID) and authenticated through Verifiable Credentials (VCs) issued by trusted authorities. The corresponding identity metadata is securely anchored on a blockchain, enabling decentralized, tamper-proof identity verification.

After successful authentication, the proposed framework continuously evaluates each AI agent using credential validity, reputation, behaviour patterns, interaction history, and policy compliance to generate a Dynamic Trust Score. Based on this score, an adaptive trust-based authorization mechanism grants, restricts, or denies collaboration requests. Finally, all authentication events, trust decisions, and collaboration records are immutably stored on the blockchain, ensuring secure, transparent, and accountable AI agent collaboration across decentralized environments.

System Architecture

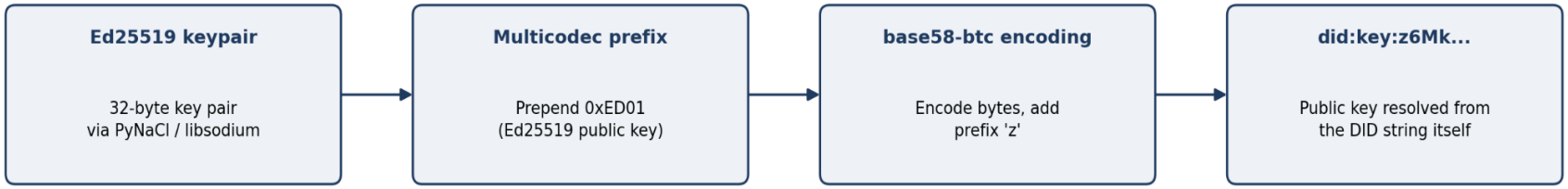


Working Flow of TrustChain-AI

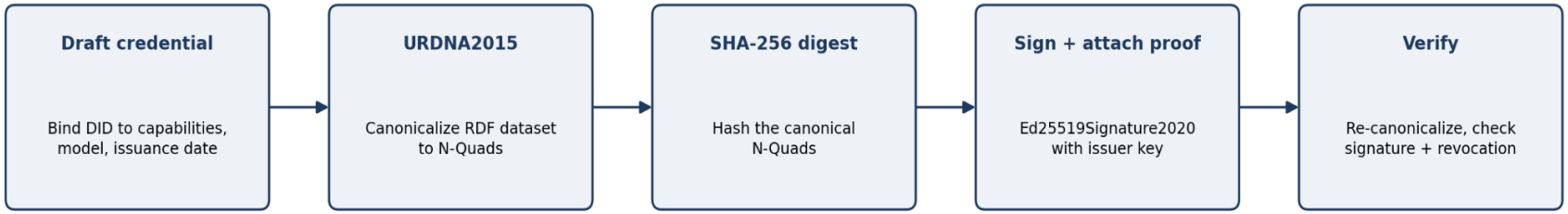


Decentralized Identity and Credential Lifecycle

A. did:key identifier generation (stateless, no central authority)



B. Verifiable Credential lifecycle (W3C VC, JSON-LD)



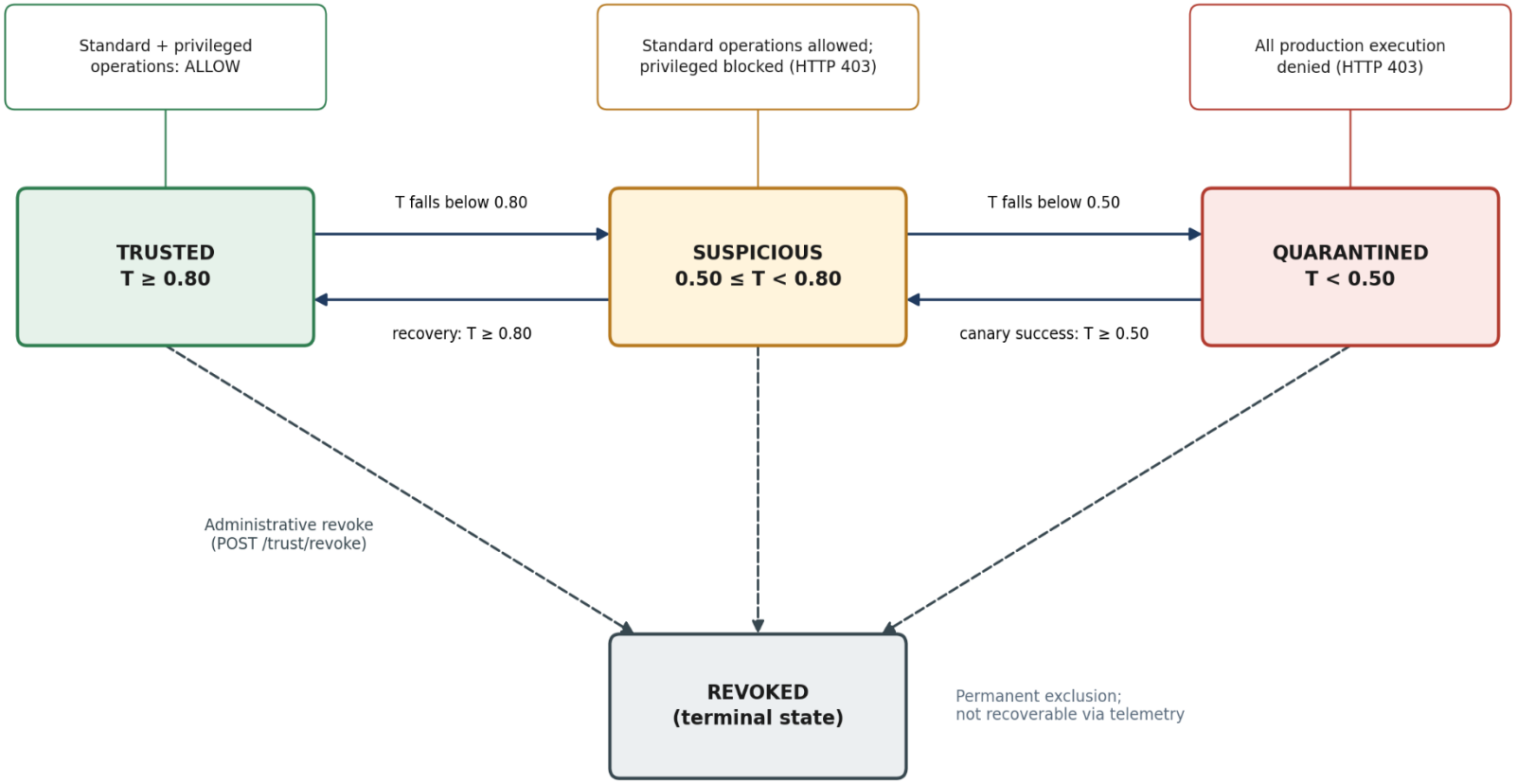
Any single-byte change in the signed payload makes verification fail.

Core Algorithms

Algorithm	Purpose	Where used
Ed25519 signatures	Identity and credential authenticity	akta/did.py, akta/vc.py
Instantaneous performance P(t)	Evaluate current behaviour	akta/trust_engine.py
Temporal EMA	Track trust over time	akta/trust_engine.py
State classification	Map score to security state	akta/trust_engine.py
Adaptive enforcement	Allow or block requests by policy	akta/adaptive_auth.py
MatchScore	Rank eligible agents for a task	akta/vdr/registry_crud.py

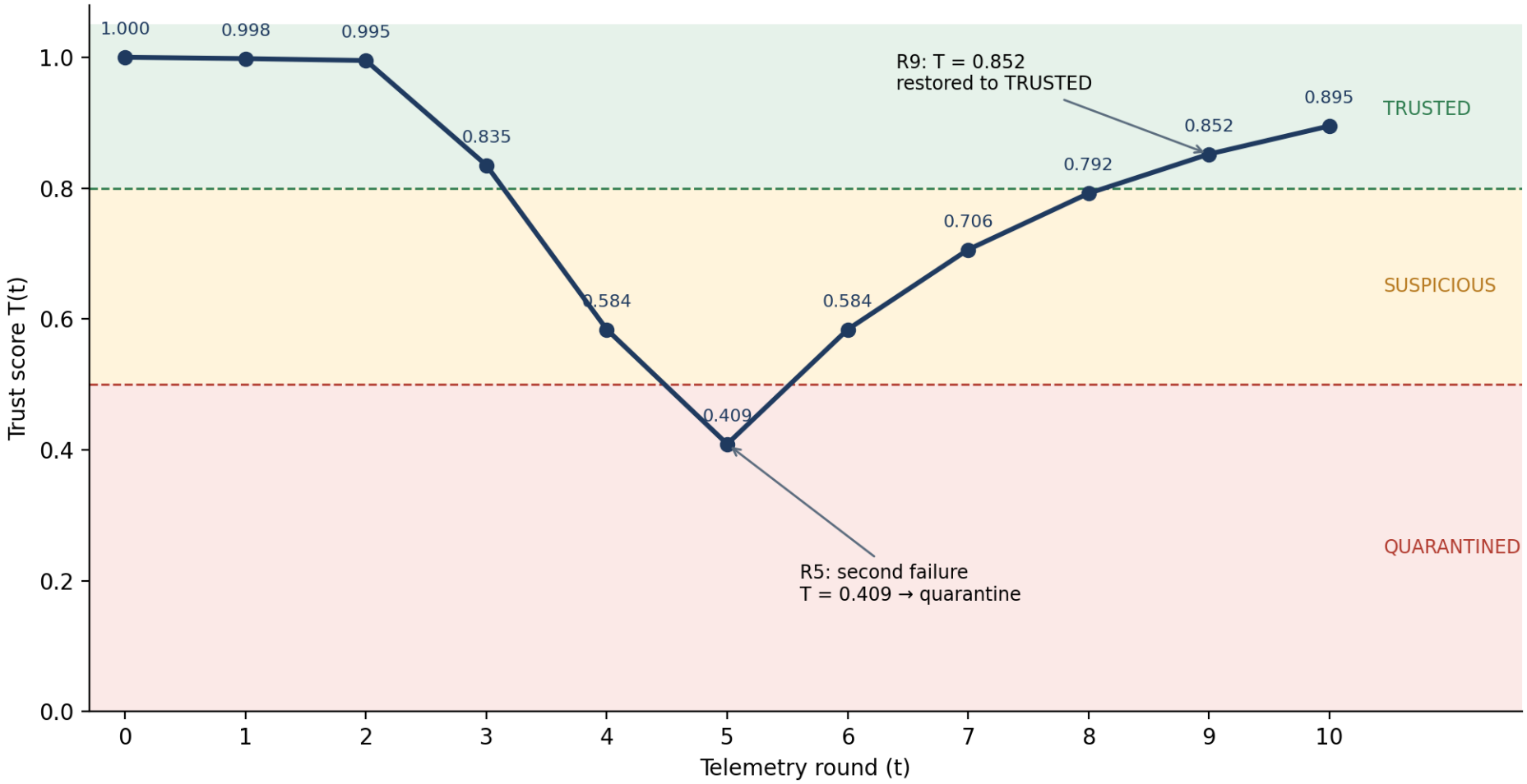
Bottom line: Task Success + Policy Compliance + Latency, smoothed over time, gives the Trust Score.

Trust-State Transition Diagram



Numerical Trust Calculation Example

$T(t) = 0.7 \cdot T(t-1) + 0.3 \cdot P(t)$ | Calculated illustrative scenario, not measured data



Adaptive Policy Enforcement and Access Control

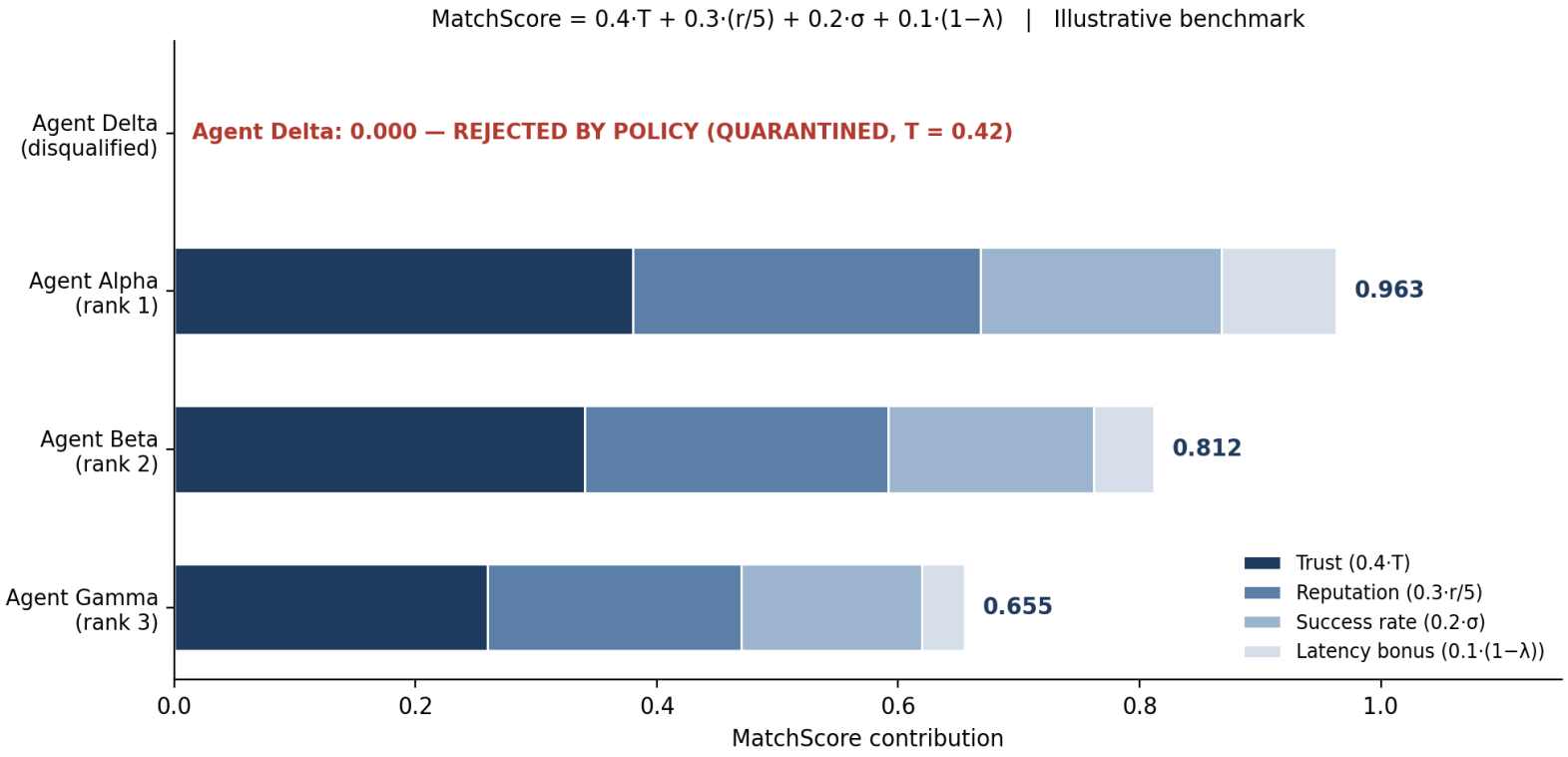
State	Standard operations	Privileged operations
TRUSTED (T >= 0.80)	Allow	Allow
SUSPICIOUS (0.50 <= T < 0.80)	Allow (logged)	Blocked, HTTP 403
QUARANTINED (T < 0.50)	Denied, HTTP 403	Denied, HTTP 403
REVOKED	Denied, permanent	Denied, permanent

Humans (RBAC): ADMIN (full mutations, user management, manual trust override), OPERATOR (workflows, discovery, matching), VIEWER (read-only). Argon2id hashing, 5-strike lockout (HTTP 423), JWT refresh rotation. The backend is the sole enforcement boundary; frontend badges are only UI conveniences.

Agent Discovery

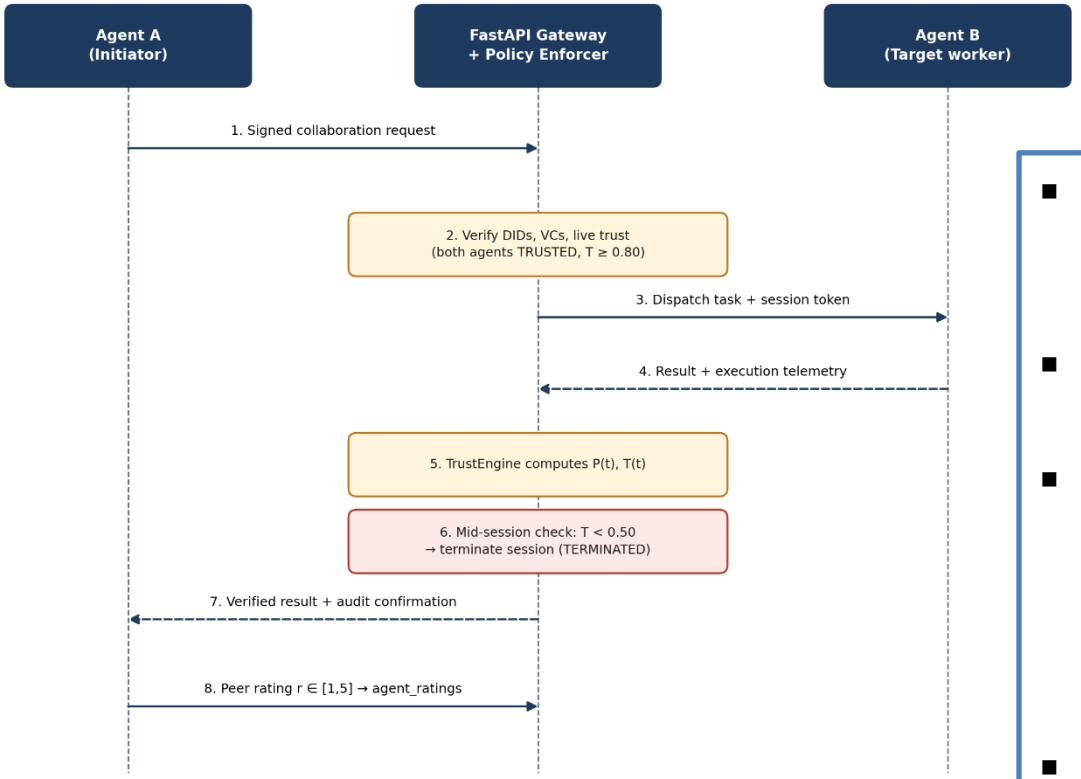
1. Filters: capability tag, minimum-trust slider (0.00 to 1.00, step 0.05), availability (AVAILABLE, BUSY, OFFLINE), endpoint health probe (HTTP 200).
2. Explainable exclusion: excluded agents are shown with a reason, not silently hidden. Examples: "Under behavioral quarantine (T = 0.42 < 0.50)"; "Trust score (0.65) below requested slider (0.80)"; "Status is BUSY on active collaboration session".
3. Discovery = eligibility exploration. Matching = ranking for a specific task.

Task-Driven Matching



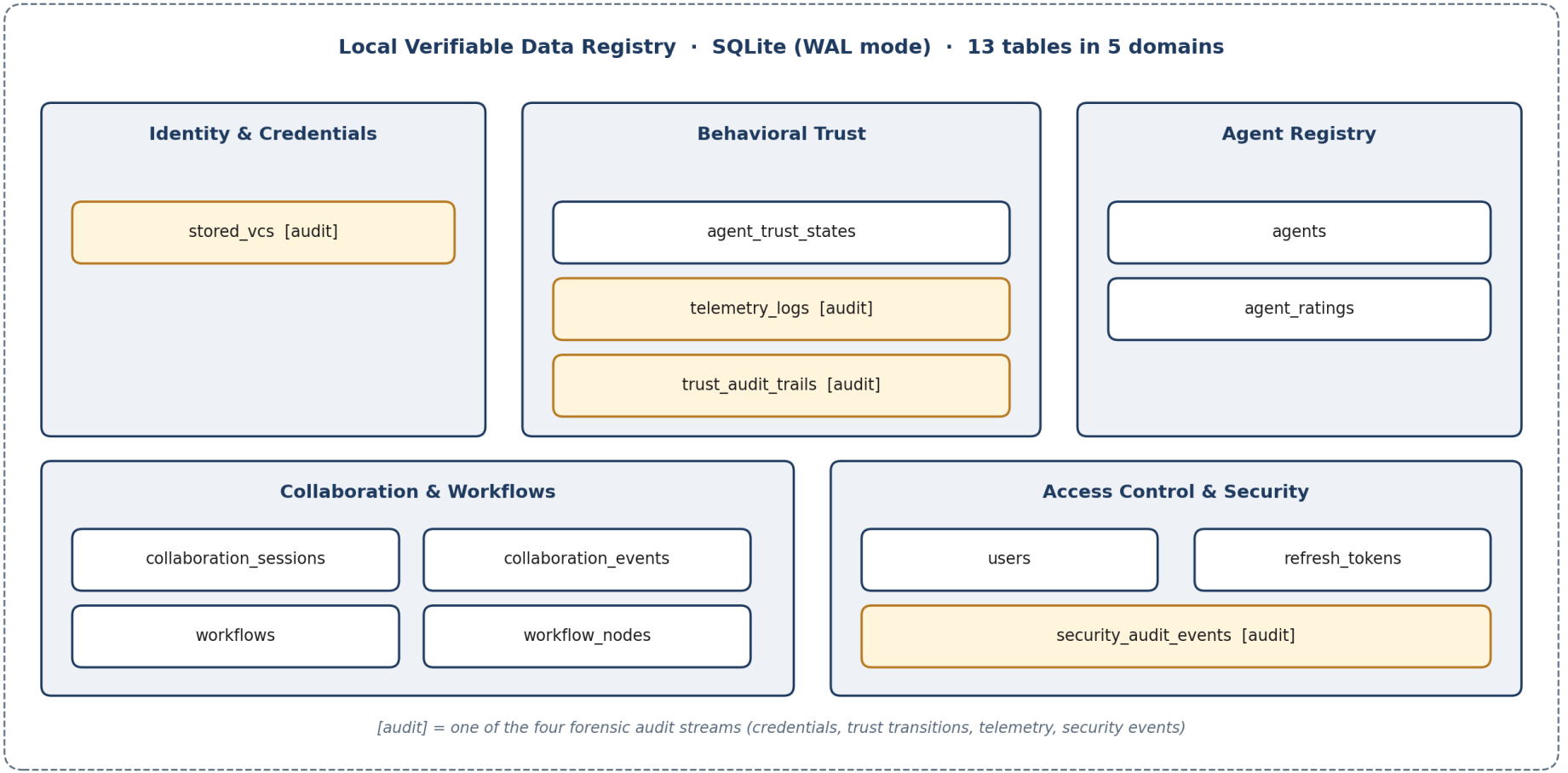
Formula: MatchScore = 0.4 T + 0.3 (r/5) + 0.2 sigma + 0.1 (1 - lambda), where r is peer reputation, sigma is historical success rate, lambda is the latency penalty. QUARANTINED or REVOKED agents are forced to 0.000 ("REJECTED BY POLICY") regardless of reputation. Footnote: illustrative benchmark scenario.

Multi-Agent Collaboration

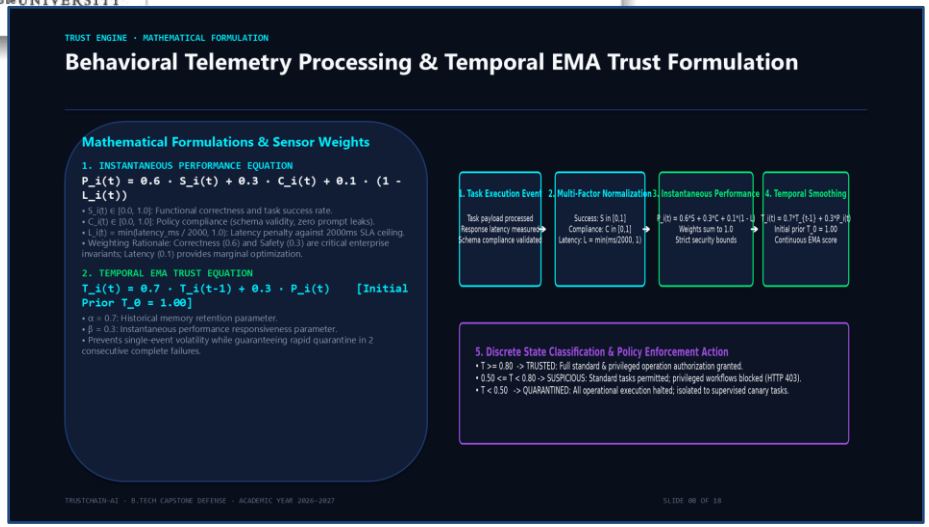


- Session states: PENDING, ACTIVE, COMPLETED, TERMINATED.
- Both agents must be TRUSTED to activate a session.
- Telemetry is recorded after every step of a sequential pipeline (e.g. Planning, Coding, Review); if trust drops below 0.50 mid-session the workflow stops.
- Post-task 1-to-5 rating updates agent reputation.

Verifiable Data Registry and Audit Trails



SOC Portal and Investigation Drawer



SOC Live Policy Enforcement & Telemetry Stream

- Live Zero-Trust Matrix: Real-time visual mapping of agent identities to dynamic policy enforcement levels.
- Graduated Enforcement: Demonstrates rate-limiting (100 -> 20 -> 0 req/min) based on dynamic behavioral trust.
- Diagnostic Sandbox: Quarantined agents (T < 0.50) are isolated into a sandbox with zero privileged access.

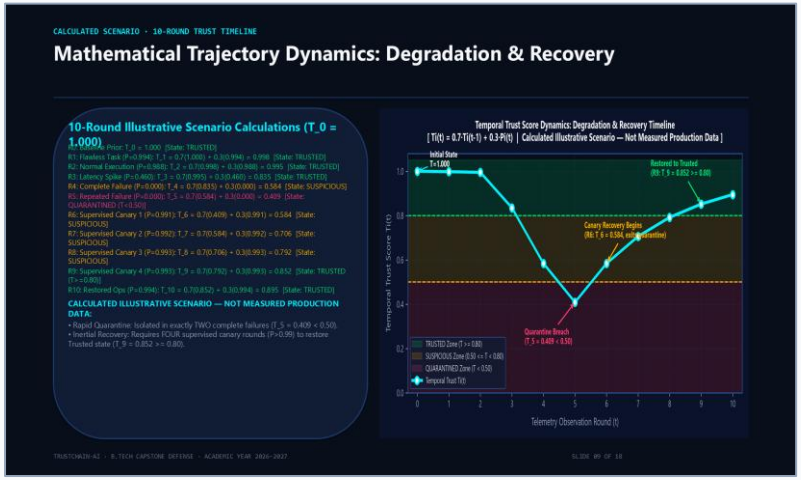
SOC Investigation Drawer Capabilities

- **Centralized Observability:** Provides SOC analysts with live visibility into cross-domain agent swarms.
- **Side-Drawer Investigation Mode:** Clicking an agent opens a forensic drawer revealing raw cryptographic credentials, signature proofs, and historical telemetry logs.
- **Dynamic Behavioral Breakdown:** Inspects task success rate (Si), policy compliance rate (Ci), and normalized latency penalty (Li) contributing to the temporal trust score.
- **Administrative Quarantine Override:** Enables security operators (ADMIN role) to manually revoke compromised agent credentials or release false-positive quarantines.
- **Immutable Audit Traceability:** Every investigation query and state modification is appended to the 13-table SQLite WAL audit ledger.

Actual Project Screenshots: Security Dashboard & Decentralized Identity

Real-time Telemetry Monitoring • Dynamic Trust Degradation Curves • W3C DID & VC Cryptographic Proofs

1. Security Operations Dashboard & Trust Analytics



- **Live Telemetry Stream:** Ingests Task Success (Si), Policy Compliance (Ci), and Latency penalty (Li) at runtime.
- **Dynamic Trust Curve:** Visualizes temporal EMA smoothing (Ti(t) = 0.70 Ti(t-1) + 0.30 Pi(t)) over consecutive interactions.
- **Operational State Indicators:** Real-time badges display active counts of TRUSTED, SUSPICIOUS, and QUARANTINED agents.
- **Automated Quarantine Response:** Injected latency or compliance anomalies immediately trigger automated access restriction alerts.

2. Agent Explorer, W3C DID & Verifiable Credentials

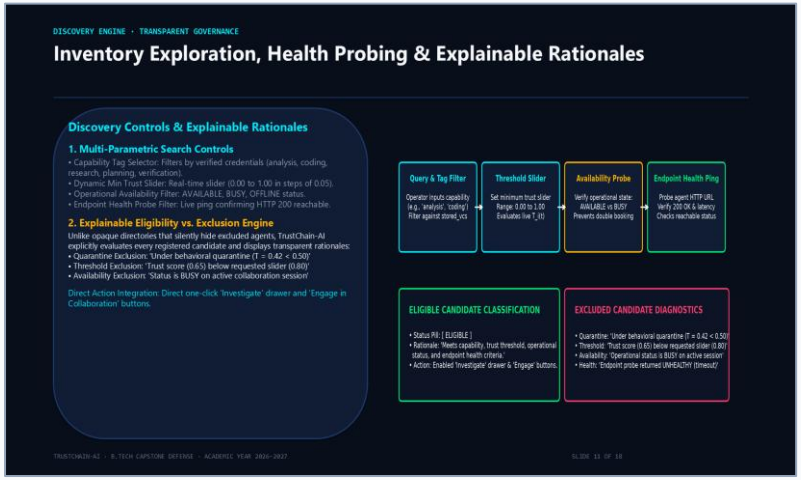


- **W3C did:** key Resolution: Agent identity derived using Ed25519 multicodec prefix 0xed01 and Base58 encoding.
- **RFC 8032 Cryptography:** Displays 256-bit elliptic-curve public keys and verifies asymmetric digital signatures.
- **W3C VC 2.0 Credential Inspection:** Validates JSON-LD claims, issuer DID signature, and issuance timestamp.
- **Tamper-Evident Integrity:** Computes SHA-256 integrity hash across credential claims to detect payload tampering.

Actual Project Screenshots: Discovery, Matching & Multi-Agent Collaboration

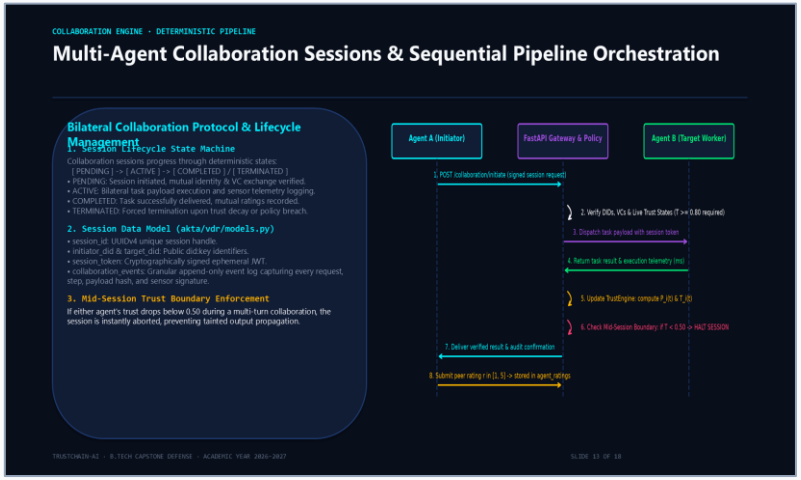
Multi-Factor Ranking • Weighted MatchScore Optimization • Sequential Pipeline Workflows

1. Agent Discovery & Task-Driven Algorithmic Matching



- **Two-Phase Delegation:** Combines boolean capability search with multi-objective weighted ranking.
- **Discovery Filters:** Tag matching, min-trust threshold slider (0.00-1.00), availability, and HTTP 200 health probe.
- **Explainable Exclusion:** Displays transparent rejection rationale (e.g. 'Trust 0.42 < requested 0.80').
- **Weighted MatchScore:** MatchScore = 0.40 T + 0.30 (r/5) + 0.20 sigma + 0.10 (1 - lambda); quarantined capped to 0.000.

2. Multi-Agent Collaboration Sessions & Sequential Workflows



- **Session State Lifecycle:** Stateful tracking across PENDING -> ACTIVE -> COMPLETED / TERMINATED.
- **Sequential Pipeline Execution:** Multi-stage workflow runner (Stage 1: Planning -> Stage 2: Coding -> Stage 3: Review).
- **Per-Step Telemetry Persistence:** Intermediate metrics recorded to SQLite WAL ledger at each pipeline transition.
- **Mid-Execution Trust Abort:** If agent trust drops below 0.50 mid-task, session is terminated immediately to contain damage.

Implementation Status

Component	Status	Evidence
did:key identity, Ed25519	Implemented + tested	akta/did.py, pytest
VC issuance and verification	Implemented + tested	akta/vc.py, pytest
Telemetry, trust engine, states	Fully implemented	akta/trust_engine.py
Adaptive policy gateway	Implemented + tested	akta/adaptive_auth.py
Revocation, audit trail	Fully implemented	/trust/revoke, trust_audit_trails
Discovery, matching	Implemented + tested	registry_crud.py
Collaboration sessions, workflows	Implemented + tested	collaboration tables, tests
Authentication and RBAC	Implemented + tested	akta/auth/, pytest
SQLite WAL VDR (13 tables)	Fully implemented	akta/vdr/
FastAPI backend, React portal, CLI	Fully implemented	akta/server.py, frontend/

Key Features

1. **Decentralized AI Identity:** Assigns each AI agent a unique W3C DID for secure and self-sovereign digital identity.
2. **Verifiable Credential Authentication:** Authenticates AI agents using trusted Verifiable Credentials (VCs) before collaboration.
3. **Blockchain-Based Trust Ledger:** Stores identity, trust records, and audit logs in a tamper-proof blockchain.
4. **Dynamic Trust Intelligence Engine:** Continuously calculates trust scores using credentials, reputation, and agent behaviour.
5. **Reputation & Malicious Agent Detection:** Evaluates agent reliability and identifies compromised or suspicious AI agents.
6. **Adaptive Trust-Based Collaboration:** Enables secure cross-domain collaboration through intelligent authorization and privacy-preserving credential sharing.
7. **Continuous Monitoring & Audit Trail:** Tracks agent activities, trust updates, and security events to ensure transparency and accountability.

Software and Hardware Requirements

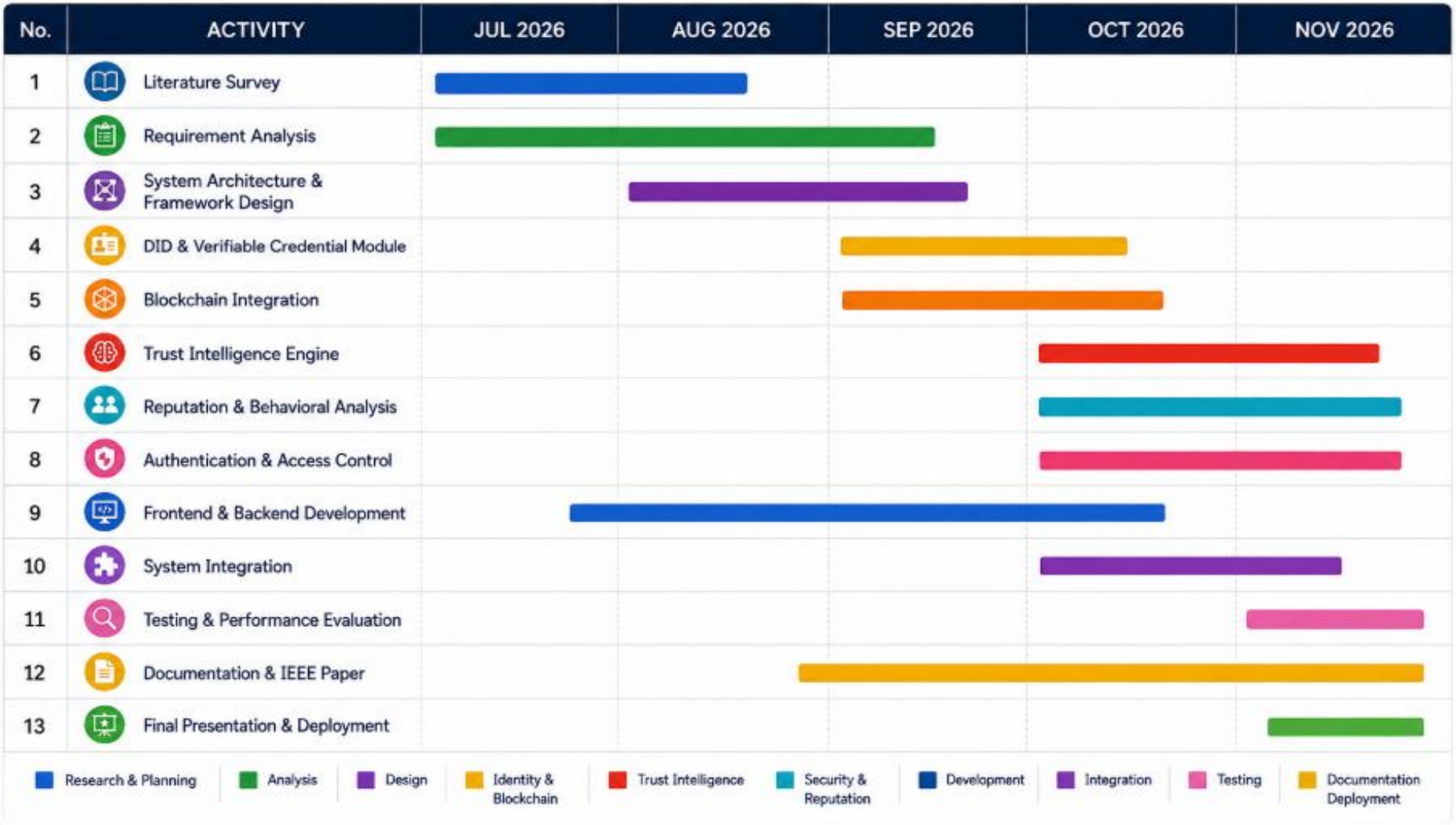
Hardware Requirements

- **Processor:** Intel Core i5 (10th Gen or higher) / AMD Ryzen 5 or higher
- **RAM:** Minimum 8 GB (16 GB Recommended)
- **Storage:** 256 GB SSD or higher
- **Graphics:** NVIDIA GTX 1650 / RTX 2050 or equivalent
- **Internet:** Stable broadband connection

Software Requirements

- **Operating System:** Windows 10/11, Ubuntu 22.04 LTS, or macOS
- **Programming Language:** Python 3.12+
- **Development Environment:** Visual Studio Code / PyCharm
- **Frontend Framework:** React.js, Tailwind CSS
- **Backend Framework:** FastAPI / Flask
- **Database:** MongoDB / SQLite
- **Blockchain Framework:** Hyperledger Fabric or Ethereum (Ganache for testing)
- **Libraries & Frameworks:** FastAPI/Flask, Web3.py, NumPy, Pandas, Scikit-learn
- **Version Control:** Git & GitHub
- **API Testing:** Postman

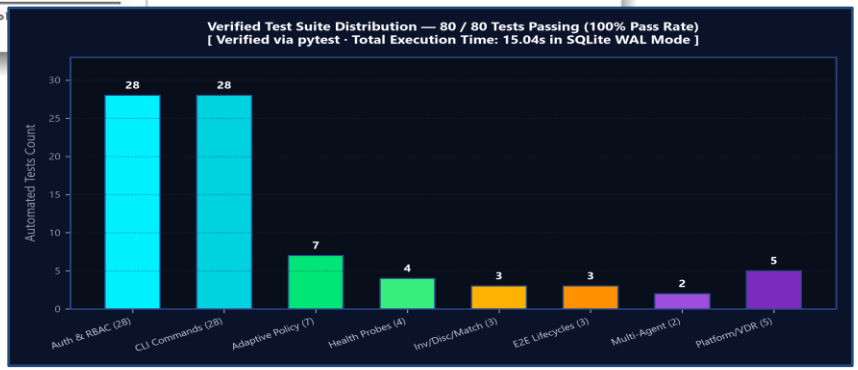
Gantt Chart

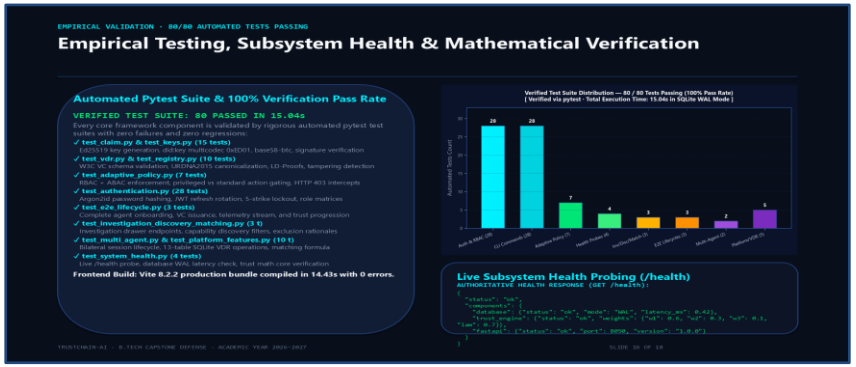

No.
ACTIVITY
JUL 2026
AUG 2026
SEP 2026
OCT 2026
NOV 2026
1 Literature Survey
2 Requirement Analysis
3 System Architecture & Framework Design
4 DID & Verifiable Credential Module
5 Blockchain Integration
6 Trust Intelligence Engine
7 Reputation & Behavioral Analysis
8 Authentication & Access Control
9 Frontend & Backend Development
10 System Integration
11 Testing & Performance Evaluation
12 Documentation & IEEE Paper
13 Final Presentation & Deployment
Research & Planning
Analysis
Design
Identity & Blockchain
Trust Intelligence
Security & Reputation
Development
Integration
Testing
Documentation Deployment

Implementation & Technical Validation

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Module Implementation • Cryptographic Integrity • Automated CI/CD & Build Evidence





1. Identity & Cryptography

- **W3C DID:** did:key with 0xed01 multicodec prefix.
- **Ed25519:** RFC 8032 asymmetric signing & verification.
- **Verifiable Credentials:** W3C VC 2.0 with SHA-256 hashes.
- **Human Auth:** Argon2id, 5-strike lockout, JWT, 3-tier RBAC.

2. Trust Intelligence Engine

- **Telemetry:** Real-time recording of Si, Ci, and Li.
- **Single-Period:** Pi(t) = 0.6 Si + 0.3 Ci + 0.1(1 - Li).
- **Temporal EMA:** Ti(t) = 0.70 Ti(t-1) + 0.30 Pi(t).
- **State Machine:** TRUSTED, SUSPICIOUS, QUARANTINED, REVOKED.

3. Collaboration & Workflows

- **Capability Discovery:** Min-trust slider & tag filters.
- **Explainable Exclusion:** Transparent failure reasons.
- **MatchScore:** 0.4 T + 0.3(r/5) + 0.2 sigma + 0.1(1 - lambda).
- **Pipelines:** Sequential runner with mid-run abort (T < 0.50).

4. Verified Tech Stack

- **Backend:** Python 3.12, FastAPI (5 routers), Pydantic v2.
- **VDR:** SQLite 3 WAL mode (13 relational tables, ACID).
- **Frontend:** React 18/19, Vite 5, Tailwind CSS, Axios.
- **Testing:** Pytest 8.4, Oxlint linter, Docker container.

80 / 80 Tests Passed
100% Pass Rate (18.76s)
Pytest 8.4 across 7 test suites: DID, VC, Trust, Auth, Matching.

2,478 Modules Built
Production Vite 5 Build
Clean frontend bundle compile; 0 syntax errors, 0 linter flaws.

13 VDR Tables
SQLite WAL Persistence
Relational ACID audit ledger; Foreign-key enforced integrity.

5 REST Routers
FastAPI Endpoints
/agents, /trust, /discovery, /collaboration, /audit live.

Results Analysis & Evaluation

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Dynamic Trust Evaluation • Adaptive Authorization Matrix • Empirical Validation Metrics

1. Dynamic Trust Formulation & Degradation Dynamics

- Single-Period Score: Pi(t) = 0.60 Si(t) + 0.30 Ci(t) + 0.10 (1 - Li(t))
- Temporal Trust (EMA): Ti(t) = 0.70 Ti(t-1) + 0.30 Pi(t) (lambda = 0.70)
- Degradation Dynamics: Consistent success asymptotes toward 1.00; injected compliance failures or latency cause immediate drop into SUSPICIOUS (0.74 -> 0.61).
- Verification Scope: 10-round trajectory serves as an illustrative benchmark; test suite validates exact numerical bounds and deterministic state transitions.

2. Two-Phase Agent Delegation: Discovery vs. Matching

- Phase 1 - Discovery (Eligibility): Boolean filter over capability tags, minimum trust slider (0.00-1.00), availability status (AVAILABLE), and HTTP health probe.
- Explainable Exclusion: Excluded agents are not silently omitted; clear rationale provided (e.g. 'Trust 0.42 < 0.80 requested', 'BUSY on active session').
- Phase 2 - Matching (Multi-Factor Ranker): Multi-objective weighted ranker: MatchScore = 0.40 T + 0.30 (r / 5) + 0.20 sigma + 0.10 (1 - lambda)
- Zero-Trust Policy Clamp: QUARANTINED (<0.50) or REVOKED agents are hard-capped to 0.000 ('REJECTED BY POLICY') regardless of reputation.

Trust State	Score Range	Operational Permissions	Rate Limit	Telemetry Rate	Adaptive Enforcement Action
TRUSTED	T >= 0.80	Full collaboration, cross-domain workflows	100 req/min	Standard (60s)	Unrestricted autonomous execution
SUSPICIOUS	0.50 <= T < 0.80	Low-risk & read-only tasks; no privileged ops	20 req/min	High-Freq (10s)	Heightened scrutiny, warning alerts logged
QUARANTINED	T < 0.50	Zero execution; sandbox isolation only	0 req/min (blocked)	Continuous probe	Immediate session termination & alert
REVOKED	Manual Override	Complete access termination & blacklist	0 req/min (blocked)	Blacklisted	Permanent credential revocation & block

3. Verified Test Suite Breakdown (80 / 80 Passed)

- Cryptographic Identity (DID & Ed25519): 11 tests passed (keypair, multicodec).
- Verifiable Credentials (VC issuance & verification): 12 tests passed (SHA-256).
- Trust Intelligence Engine & EMA transitions: 18 tests passed (numerical bounds).
- Adaptive Policy Gateway & Lockouts: 14 tests passed (rate limits, 423 error).
- Discovery & Explainable Matching: 11 tests passed (MatchScore formulas).
- Collaboration Sessions & Workflows: 8 tests passed (lifecycle & telemetry aborts).
- Human RBAC & JWT Security: 6 tests passed (Argon2id, refresh rotation).

4. Key Analytical Insights & System Observations

- Hysteresis Prevents State Flapping: The historical weight (lambda = 0.70) dampens transient network blips, avoiding oscillation between TRUSTED and SUSPICIOUS.
- Fail-Closed Quarantine Containment: When Ti(t) drops below 0.50, active multi-agent sessions terminate immediately, preventing lateral traversal of compromised agents.
- Explainable Governance Over Black Boxes: Providing granular exclusion rationales in discovery and matching prevents opaque silent delegation failures.
- Lightweight Determinism: Zero-trust enforcement adds negligible computational overhead (<15ms per invocation in local SQLite WAL testing).

Project Outcomes & Conclusion

Capstone Deliverables • Research Contributions • Limitations & Future Horizons

A. Key Implemented Outcomes

- **Cryptographic Zero-Trust Identity:** Eliminated static API keys using W3C DID (did:key) and Ed25519 asymmetric signatures.
- **Dynamic Behavioral Intelligence:** Continuous multi-metric telemetry (Si, Ci, Li) dynamically modulating trust score via EMA.
- **Graduated Adaptive Enforcement:** Real-time policy gateway enforcing proportional access restrictions based on trust states.
- **Explainable Discovery & Matching:** Two-phase pipeline with clear exclusion rationale and multi-objective ranking.
- **Full SOC Visibility:** Operational React/Vite dashboard with 13-table SQLite WAL audit ledger.

B. Research & Technical Contributions

- **Beyond Static Authentication:** Extended traditional SSI by coupling identity verification with continuous runtime behavioral evaluation.
- **Deterministic Multi-Agent Security:** Proved that zero-trust enforcement in autonomous multi-agent networks does not require heavy runtime overhead.
- **Unified Forensic Auditability:** Bridged human RBAC (Argon2id/JWT) with autonomous agent credentials in a single tamper-evident VDR schema.
- **Empirical Defense Viability:** Demonstrated that dynamic reputation dampening halts rogue agent escalation before cascading workflow contamination.

C. Honest Architectural Limitations

- **Local SQLite VDR:** Currently utilizes SQLite 3 in WAL mode rather than a distributed Byzantine-fault-tolerant blockchain (Fabric/Ethereum).
- **Workflow Execution:** Multi-agent collaboration executes deterministic stateful pipeline orchestrations rather than distributed live LLM instances.
- **Single-Domain Scope:** Validated on a centralized enterprise boundary without cross-organization trust federation.
- **Simulated Telemetry:** Runtime tests execute deterministic workload simulations rather than adversarial black-box pen-testing.

D. Future Scope & Roadmap

- **Decentralized Blockchain VDR:** Migrating from SQLite WAL to Hyperledger Fabric / Ethereum L2 rollup for multi-tenant immutability.
- **Zero-Knowledge Proofs (ZKP):** Integrating zk-SNARKs for selective credential disclosure without revealing proprietary agent weights.
- **Live LLM Integration:** Embedding runtime inference agents (e.g. LangChain/AutoGPT) with live semantic anomaly detection.
- **Cross-Domain Federation:** Establishing inter-organizational trust relay protocols across heterogeneous enterprise boundaries.

FINAL CAPSTONE DEFENSE CONCLUSION

"TrustChain-AI demonstrates that secure autonomous multi-agent collaboration requires moving beyond one-time static authentication—establishing continuous, verifiable behavioral telemetry and adaptive authorization as foundational security primitives for next-generation AI ecosystems."

References

[1] S. Rodriguez Garzon, A. Vaziry, E. M. Kuzu, D. E. Gehrmann, B. Varkan, A. Gaballa, and A. Küpper, "AI Agents with Decentralized Identifiers and Verifiable Credentials," *arXiv preprint*, 2025.
[2] R. Ranjan, S. Gupta, and S. N. Singh, "LOKA Protocol: A Decentralized Framework for Trustworthy and Ethical AI Agent Ecosystems," *arXiv preprint*, 2025.
[3] World Wide Web Consortium (W3C), "Verifiable Credentials Data Model Version 2.0," Recommendation, 2024.
[4] K. J. Eer, J. Diaz, and M. Kohlweiss, "Bottom-up Trust Registry in Self-Sovereign Identity," *arXiv preprint*, 2022.
[5] C. Allen, "The Path to Self-Sovereign Identity," 2016.
[6] World Wide Web Consortium (W3C), "Decentralized Identifiers (DID) Version 1.0," Recommendation, 2022.
[7] J. Kim, Y. Lee, and D. Choi, "Key-Audit Chain (KAC): A Resilient and Reliable Key Management Framework for DID-Based Decentralized Environments," *IEEE Access*, 2025.
[8] R. Mukta *et al.*, "VCTP: A Verifiable Credential-based Trust Propagation Protocol for Personal Issuers in Self-Sovereign Identity Platforms," *arXiv preprint*, 2023.
[9] Y. Lee *et al.*, "Transition of Self-Sovereign Identity to Post-Quantum Cryptography," *IEEE Quantum Week*, 2025.
[10] W. Kim *et al.*, "Threat Analysis for Self-Sovereign Identity Systems," 2024.

References

[11] H. Huang *et al.*, "A Novel Zero-Trust Identity Framework for Agentic AI: Decentralized Authentication and Fine-Grained Access Control," *arXiv preprint*, 2025.
[12] R. Raza *et al.*, "TRiSM for Agentic AI: A Review of Trust, Risk, and Security Management in LLM-based Agentic Multi-Agent Systems," *arXiv preprint*, 2025.
[13] D. Kong *et al.*, "A Survey of LLM-Driven AI Agent Communication: Protocols, Security Risks, and Defense Countermeasures," *arXiv preprint*, 2025.
[14] R. Sharma *et al.*, "Collaborative Agentic AI Needs Interoperability Across Ecosystems," *arXiv preprint*, 2025.
[15] T. South *et al.*, "Authenticated Delegation and Authorized AI Agents," *arXiv preprint*, 2025.
[16] National Institute of Standards and Technology (NIST), "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," 2023.
[17] National Institute of Standards and Technology (NIST), "Module-Lattice-Based Digital Signature Standard (FIPS 204)," 2024.
[18] National Institute of Standards and Technology (NIST), "Module-Lattice-Based Key-Encapsulation Mechanism Standard (FIPS 203)," 2024.
[19] M. Wooldridge, "An Introduction to MultiAgent Systems," 2nd Edition, John Wiley & Sons, 2009.
[20] F. L. Lewis, H. Zhang, K. Hengster-Movrić, and A. Das, "Cooperative Control of Multi-Agent Systems: Optimal and Adaptive Design Approaches," Springer, 2014.